

NanoGT Match Report: Glycogen storage disease type Ia

Disease: Glycogen storage disease type Ia (ORPHA:79258)

Primary gene: G6PC

Gene CDS: 1071 bp

Inheritance: Autosomal recessive

Target tissues: liver, kidney

Top 5 GT Precedent Matches

Rank	Program	Vector	Score	Confidence	Approval
1	SRP-9001	AAV9	7.4/10	[YELLOW] Medium	approved
2	ST-920	AAV2/6	7.4/10	[YELLOW] Medium	phase1/2
3	RGX-121	AAV9	7.3/10	[YELLOW] Medium	phase3
4	Hemgenix	AAV5	7.2/10	[YELLOW] Medium	approved
5	Roctavian	AAV5	7.2/10	[YELLOW] Medium	approved

Match #1: SRP-9001

Precedent disease: Duchenne muscular dystrophy

Vector: AAV9

Tissue target: muscle

Composite score: 7.4 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	1.00	2.0	Vector naturally reaches disease target tissue
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	2.00	2.0	Same or related biological pathway
Inheritance compatibility	0.70	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	1.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	1.20	2.0	Can GT be given before irreversible damage?

Dimension	Score	Max	What it measures
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.80	1.0	Immunological protection of target tissue
Promoter availability	1.00	1.0	Validated tissue-specific promoters exist
Route of administration	1.00	1.0	Established delivery route to target tissue
TOTAL (normalised)	7.44	10.0	Raw sum / 18 × 10

Rationale

- Gene CDS 1071bp / cargo 4700bp (23% utilized)
- Vector tropism overlaps liver, but precedent target is muscle
- Both membrane proteins
- LOF inheritance — compatible for gene replacement
- Pathway match: myopathy
- Approval status: approved
- Vector immunogenicity (AAV9): high (~22%) — substantial patient exclusion expected; immunodepletion protocols may be needed
- Moderate-to-narrow window — early childhood onset; newborn screening integration would significantly improve outcomes
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: moderate-high privilege — tolerogenic microenvironment (Kupffer cells, IL-10, PD-L1)
- Promoter availability: ApoE/hAAT, TBG, transthyretin, albumin — extensively validated; used in Hemgenix, Roctavian, DTX301
- Route of administration: IV systemic — established, minimally invasive; used in all hepatic GT programs

Match #2: ST-920

Precedent disease: Fabry disease

Vector: AAV2/6

Tissue target: liver

Composite score: 7.4 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue

Dimension	Score	Max	What it measures
Protein class	0.50	2.0	Same se-creted/lysosomal/membrane/intracellular class
Pathway similarity	2.00	2.0	Same or related biological pathway
Inheritance compatibility	0.70	1.0	AR/XL loss-of-function pattern match
Approval precedent	0.50	1.0	Regulatory approval / trial stage
Immunogenicity	1.50	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	1.20	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.80	1.0	Immunological protection of target tissue
Promoter availability	1.00	1.0	Validated tissue-specific promoters exist
Route of administration	1.00	1.0	Established delivery route to target tissue
TOTAL (normalised)	7.44	10.0	Raw sum / 18 × 10

Rationale

- Gene CDS 1071bp / cargo 4700bp (23% utilized)
- Vector tropism plus precedent target match: liver
- Protein class mismatch
- LOF inheritance — compatible for gene replacement
- Pathway match: lysosomal_storage
- Approval status: phase1/2
- Vector immunogenicity (AAV2/6): moderate (~17%) — significant proportion may require pre-screening or exclusion
- Moderate-to-narrow window — early childhood onset; newborn screening integration would significantly improve outcomes
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: moderate-high privilege — tolerogenic microenvironment (Kupffer cells, IL-10, PD-L1)
- Promoter availability: ApoE/hAAT, TBG, transthyretin, albumin — extensively validated; used in Hemgenix, Roctavian, DTX301
- Route of administration: IV systemic — established, minimally invasive; used in all hepatic GT programs

Match #3: RGX-121

Precedent disease: Mucopolysaccharidosis type II

Vector: AAV9

Tissue target: CNS/liver

Composite score: 7.3 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	0.50	2.0	Same se-creted/lysosomal/membrane/intracellular class
Pathway similarity	2.00	2.0	Same or related biological pathway
Inheritance compatibility	0.70	1.0	AR/XL loss-of-function pattern match
Approval precedent	0.80	1.0	Regulatory approval / trial stage
Immunogenicity	1.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	1.20	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.80	1.0	Immunological protection of target tissue
Promoter availability	1.00	1.0	Validated tissue-specific promoters exist
Route of administration	1.00	1.0	Established delivery route to target tissue
TOTAL (normalised)	7.33	10.0	Raw sum / 18 × 10

Rationale

- Gene CDS 1071bp / cargo 4700bp (23% utilized)
- Vector tropism plus precedent target match: liver
- Protein class mismatch
- LOF inheritance — compatible for gene replacement
- Pathway match: lysosomal_storage
- Approval status: phase3
- Vector immunogenicity (AAV9): high (~22%) — substantial patient exclusion expected; immunodepletion protocols may be needed
- Moderate-to-narrow window — early childhood onset; newborn screening integration would significantly improve outcomes

- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: moderate-high privilege — tolerogenic microenvironment (Kupffer cells, IL-10, PD-L1)
- Promoter availability: ApoE/hAAT, TBG, transthyretin, albumin — extensively validated; used in Hemgenix, Roctavian, DTX301
- Route of administration: IV systemic — established, minimally invasive; used in all hepatic GT programs

Match #4: Hemgenix

Precedent disease: Hemophilia B

Vector: AAV5

Tissue target: liver

Composite score: 7.2 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	0.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	0.50	2.0	Same or related biological pathway
Inheritance compatibility	0.70	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	1.20	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.80	1.0	Immunological protection of target tissue
Promoter availability	1.00	1.0	Validated tissue-specific promoters exist
Route of administration	1.00	1.0	Established delivery route to target tissue
TOTAL (normalised)	7.17	10.0	Raw sum / 18×10

Rationale

- Gene CDS 1071bp / cargo 4700bp (23% utilized)
- Vector tropism plus precedent target match: liver
- Protein class mismatch
- LOF inheritance — compatible for gene replacement
- Different pathway (glycogen_storage vs coagulation)
- Approval status: approved
- Vector immunogenicity (AAV5): low (~9%) — most patients eligible; minimal screening burden
- Moderate-to-narrow window — early childhood onset; newborn screening integration would significantly improve outcomes
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: moderate-high privilege — tolerogenic microenvironment (Kupffer cells, IL-10, PD-L1)
- Promoter availability: ApoE/hAAT, TBG, transthyretin, albumin — extensively validated; used in Hemgenix, Roctavian, DTX301
- Route of administration: IV systemic — established, minimally invasive; used in all hepatic GT programs

Match #5: Roctavian

Precedent disease: Hemophilia A

Vector: AAV5

Tissue target: liver

Composite score: 7.2 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	0.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	0.50	2.0	Same or related biological pathway
Inheritance compatibility	0.70	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	1.20	2.0	Can GT be given before irreversible damage?

Dimension	Score	Max	What it measures
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.80	1.0	Immunological protection of target tissue
Promoter availability	1.00	1.0	Validated tissue-specific promoters exist
Route of administration	1.00	1.0	Established delivery route to target tissue
TOTAL (normalised)	7.17	10.0	Raw sum / 18×10

Rationale

- Gene CDS 1071bp / cargo 4700bp (23% utilized)
- Vector tropism plus precedent target match: liver
- Protein class mismatch
- LOF inheritance — compatible for gene replacement
- Different pathway (glycogen_storage vs coagulation)
- Approval status: approved
- Vector immunogenicity (AAV5): low (~9%) — most patients eligible; minimal screening burden
- Moderate-to-narrow window — early childhood onset; newborn screening integration would significantly improve outcomes
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: moderate-high privilege — tolerogenic microenvironment (Kupffer cells, IL-10, PD-L1)
- Promoter availability: ApoE/hAAT, TBG, transthyretin, albumin — extensively validated; used in Hemgenix, Roctavian, DTX301
- Route of administration: IV systemic — established, minimally invasive; used in all hepatic GT programs